An audit of the immrep25 unseen-peptide benchmark

Signal-to-noise of CDR3 homology and V/J chain pairing across the positive set

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July 3, 2026

Abstract

We ask whether the *positive* TCR–peptide assignments in the immrep25 unseen-peptide benchmark carry epitope-specific signal, or only noise and publicity. Two biologically-grounded probes—within- vs. between-epitope CDR3 *homology* and per-epitope V/J *chain pairing*—are calibrated against cohorts of known quality (TCRvdb experimentally verified / refuted; VDJdb high- / low-reference-support) and a generation-probability–matched OLGA random floor. On both probes and both chains the immrep25 positives sit at the noise floor, indistinguishable from the pgen-matched random control; their only measurable homology is fully explained by public TCRs. We conclude the positive labels are not backed by convergent-TCR biology.

1 Data and design

Cohorts. Every cohort is reduced to a common per-chain schema (one CDR3 with its V/J and epitope) and a paired schema (cdr3$\alpha, V\alpha, J\alpha,$ cdr3$\beta, V\beta, J\beta$, epitope). The six cohorts span the expected signal-to-noise (S/N) range:

cohort	definition	role
TCRvdb true	TCRvdb, $p_{\mathrm{adj}} < 10^{-5}$	experimentally verified (top)
VDJdb HQ	VDJdb human MHC-I, ≥ 2 references per (TCR, epitope)	strongly supported
VDJdb LQ	VDJdb human MHC-I, single reference	weakly supported
TCRvdb false	TCRvdb, $p_{\mathrm{adj}} \geq 10^{-5}$	experimentally refuted
immrep25 pos	immrep25 `label`=1	**system under test**
OLGA random	P_{gen}-matched to immrep25 pos	pure-noise floor

Cohorts draw on VDJdb [1], the functionally-validated TCRvdb/MATCHMAKERS database [2], the immrep25 unseen-peptide benchmark [3], and an OLGA control built from the TCR generation model of Murugan et al. [4]. The immrep25 test set covers 20 nine-mer peptides that are *absent from VDJdb*, with 50 positives (and 450 negatives) each; only positives are audited. TCRvdb true/false are dominated by the same two epitopes (YLQPRTFLL, GLCTLVAML), giving an epitope-matched verified-vs-refuted contrast. Because the number of epitopes itself inflates S/N, all cross-cohort comparisons use a *matched design*: $E = 2$ epitopes $\times$ $K = 30$ TCRs, bootstrapped (the two-epitope cap is imposed by TCRvdb), with permutation nulls making every metric dimensionless.

Pipeline.

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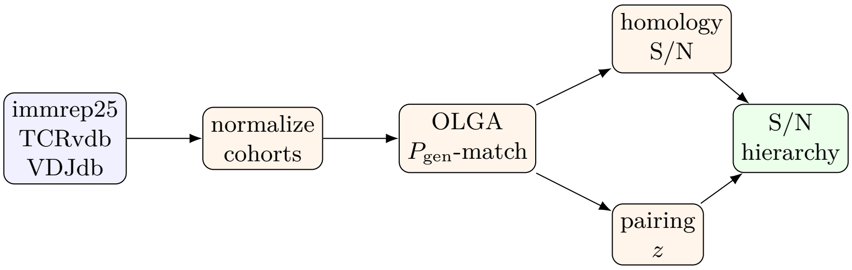

2 Homology test

Metric. For a single chain, TCRs are grouped by epitope. Only equal-length CDR3s can share a Hamming distance, so pairs are counted within length bins. With N_e TCRs for epitope e in a cohort of N,

$$P_{\text{self}} = \sum_e \binom{N_e}{2}, \qquad P_{\text{nonself}} = \binom{N}{2} - P_{\text{self}},$$

(over equal-length pairs) and, writing $m_{\le d}/l_{\le d}$ for within-/between-epitope pairs at Hamming $\le d$,

$$\text{S/N}(d) = \frac{m_{\le d}/P_{\text{self}}}{l_{\le d}/P_{\text{nonself}}}.$$

The comparison is of *rates*: non-self pairs vastly outnumber self pairs, so raw counts are not comparable—only after normalising by P_{self} and P_{nonself} does the enrichment appear. d=0 isolates publicity (identical CDR3s); d=1 is the substitution-neighbour convergence signal exploited by curated specificity databases [1]; signal $\Rightarrow$ S/N $\gg 1$, noise $\Rightarrow$ S/N ≈ 1.

Result. Figure 1 places the cohorts on the S/N axis at the matched design. The ordering is VDJdb-HQ > TCRvdb-true $\gtrsim$ TCRvdb-false > VDJdb-LQ $\gg$ immrep25 > OLGA. On TRβ the verified cohorts reach S/N = 285.33 (VDJdb-HQ) and 33.42 (TCRvdb-true), whereas immrep25 positives give only 2.54 and the P_{gen}-matched OLGA floor is 1.04. The same picture holds on TRα (immrep25 3.43 vs. OLGA 1.04).

By-epitope breakdown. To show what drives those intervals, Fig. 2 plots the within-epitope neighbour rate for every epitope with $n \ge 50$ (one point per epitope); the vertical gap to each cohort's non-self rate (red bar) is the S/N. immrep25's swarm hugs its non-self line, whereas the verified cohorts sit orders of magnitude above. Table 1 lists all 20 immrep25 epitopes: in total they contribute only 54 (TRβ) / 53 (TRα) within-epitope Hamming-1 neighbours — versus 0 / 1 for the P_{gen}-matched OLGA control and tens of thousands for the VDJdb cohorts — and even that handful is concentrated in a few epitopes (e.g. YLFNADIWI) and is entirely public (§4).

Graph view. The S/N ratio is the within- vs. between-epitope edge-density of the homology graph (nodes=CDR3, edges=Hamming≤1). Figure 3 renders it (graphviz `sfdp` layout): the verified cohorts form dense epitope-coloured cliques, while immrep25 and OLGA are almost edgeless point clouds.

3 Pairing test

Metric. Genome-wide α/β pairing is close to random [5], but given an epitope the V/J usage is constrained. Two signals are scored against permutation nulls, per cohort at the matched design:

- **gene-usage bias**: $\overline{\text{KL}} = \sum_e w_e \, \text{KL}(P_e(\text{gene}) \,\|\, Q(\text{gene}))$ over axes $V\alpha, J\alpha, V\beta, J\beta$, with P_e the within-epitope gene distribution and Q the cohort background; null = shuffle epitope labels.

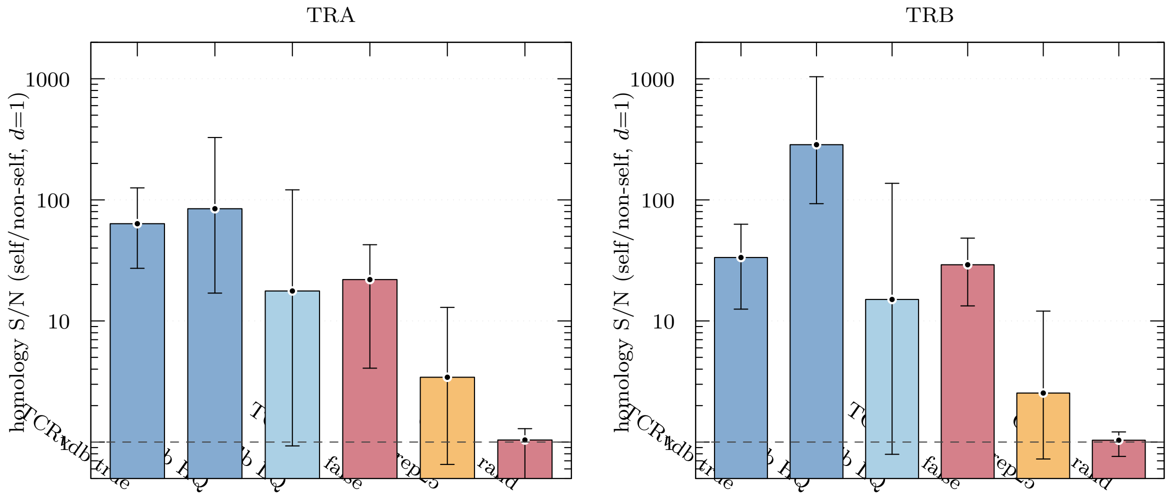

Figure 1: Homology S/N (d=1) per cohort, TRα and TRβ, matched design (E=2, K=30, 200 bootstraps; bars = median, whiskers = 95% CI). The whiskers are bootstrap confidence intervals over resampled epitopes *and* TCRs, so their width mainly reflects between-epitope variation (each bootstrap draws E=2 epitopes; per-epitope n=50 for immrep25). Dashed line: noise floor S/N=1. immrep25 positives sit just above the OLGA random floor and far below every quality cohort.

- **inter-chain MI**: $\bar{I} = \sum_e w_e \, I((V\alpha, J\alpha); (V\beta, J\beta))$ within epitope (Miller–Madow corrected); null = shuffle the $\alpha \leftrightarrow \beta$ pairing within each epitope (keeps marginals, destroys coupling).

Both are reported as $z = (\text{obs} - \mu_{\text{null}})/\sigma_{\text{null}}$; signal $\Rightarrow z \gg 0$, random $\Rightarrow z \approx 0$.

Result. Figure 4 shows both z-scores. The verified cohorts are strongly non-random (VDJdb-HQ gene-bias $z = 21.44$), while immrep25 positives are at the floor: gene-usage bias $z = 2.62$ and, decisively, an inter-chain coupling of $z = -0.06$—i.e. *no* α–β coupling beyond marginals, exactly the random-pairing expectation shared with OLGA. (The large raw MI at K=30 is finite-sample bias; the pairing-shuffle null removes it.)

4 Publicity control

The one alternative to "pure noise" is *publicity*: immrep25 might reuse public TCRs whose chance homology mimics epitope signal. Although only 0.3% of positives share *both* chains exactly with VDJdb, allowing a single substitution and either chain, 74.50% of immrep25 positives are within Hamming ≤ 1 of a known VDJdb/TCRvdb TCR (61.50% on α, 31.20% on β)—the α chain being especially germline-public. Consistently, the positives are enriched for high generation probability relative to raw OLGA sampling (Fig. 5), i.e. they are "easy-to-generate", public-leaning sequences; this is exactly why the random control must be P_{gen}-matched.

Removing public positives collapses the (already weak) homology to the noise floor: S/N falls from 2.54 to 1.00 on TRβ and from 3.43 to 1.24 on TRα (Fig. 6). The residual signal was therefore publicity, not epitope-specific convergence.

5 Conclusion

Across both probes and both chains the calibration cohorts recover the expected order, and the immrep25 positives fall at the bottom, on top of the P_{gen}-matched random floor (Fig. 7). Their only measurable homology is contributed by public TCRs and vanishes when those are removed;

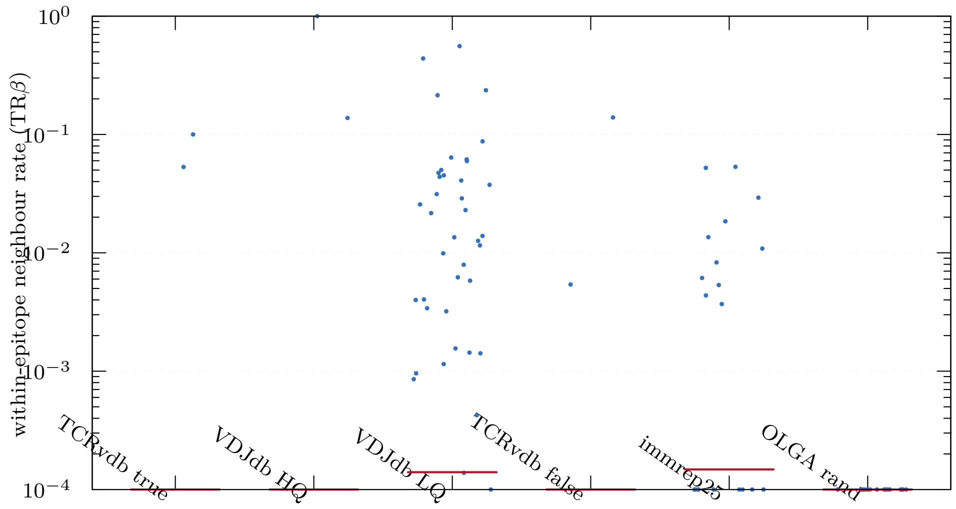

Figure 2: Per-epitope within-epitope neighbour rate (TRβ, Hamming$\leq$ 1); each point is one epitope ($n \geq 50$). Red bars = cohort pooled cross-epitope (non-self) rate; the vertical gap is the homology S/N.

their chain pairing shows no coupling beyond marginals. We therefore find *no evidence of epitope-specific signal—in homology or chain pairing—among the immrep25 positives beyond noise and publicity.*

Caveats. TCRvdb true/false separate less than expected because p_{adj} scores differential enrichment in one screen rather than absence of specificity, so "false" still contains epitope-associated TCRs; the matched design is capped at E=2 epitopes by TCRvdb. Neither affects the immrep25≈OLGA result, which is established at immrep25's own 20-epitope design as well. Reproduce with `python run_audit.py` then `make`.

References

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[3] Eve Richardson, Yannick Jurriaan Maria Aarts, John A. Altin, Coos A. B. Baakman, Philip Bradley, Binbin Chen, Joakim Clifford, Manjima Dhar, Danielle Diepenbroek, Ethan Fast, Ragul Gowthaman, Jieling He, Vadim Karnaukhov, Dario F. Marzella, Pieter Meysman, Morten Nielsen, Jonas Birkelund Nilsson, Sebastian Nymann Deleuran, Farzaneh M. Parizi,

epitope	n	β nbrs	α nbrs
`YLFNADIWI`	50	20	1
`MTDYDYLEV`	50	13	3
`ILHTHVPEV`	50	5	7
`TVYPYGTSL`	50	5	2
`YMFYDGYDV`	50	3	2
`REDDYSVWL`	50	2	1
`SQFNWTIYL`	50	2	1
`GEVLACYAL`	50	1	1
`FLDCKSYIL`	50	1	4
`FEAQPGALL`	50	1	3
`SEESAFYVL`	50	1	2
`ILMHATYFL`	50	0	1
`YEDGVIFYL`	50	0	5
`MENWSALEL`	50	0	0
`HLDDYPYLM`	50	0	1
`KLYPFLWFA`	50	0	11
`YENGSTPVL`	50	0	1
`GENALTYAL`	50	0	3
`MEMPDYLLL`	50	0	3
`YESYIPGAL`	50	0	1
immrep25 total	1000	54	53
OLGA control	999	0	1

Table 1: immrep25 positives by epitope (n TCRs) and their count of within-epitope Hamming-1 CDR3 neighbours per chain, versus the OLGA control. Sorted by TRβ.

Aurelien Pelissier, Brian G. Pierce, Maria Rodriguez Martinez, Dona Roran A R, Shayana Saravanakumar, Yanjun Shao, Nils Smit, Max Van Houcke, Gian Marco Visani, Yat-Tsai Richie Wan, Xiaowo Wang, Lawson Woods, Sander Wuyts, Chengkai Xiao, Li C. Xue, Justin Barton, Matthew Noakes, Damon H. May, Bjoern Peters, and IMMREP25 Participant Consortium. Immrep25: Unseen peptides. *bioRxiv*, 2026.

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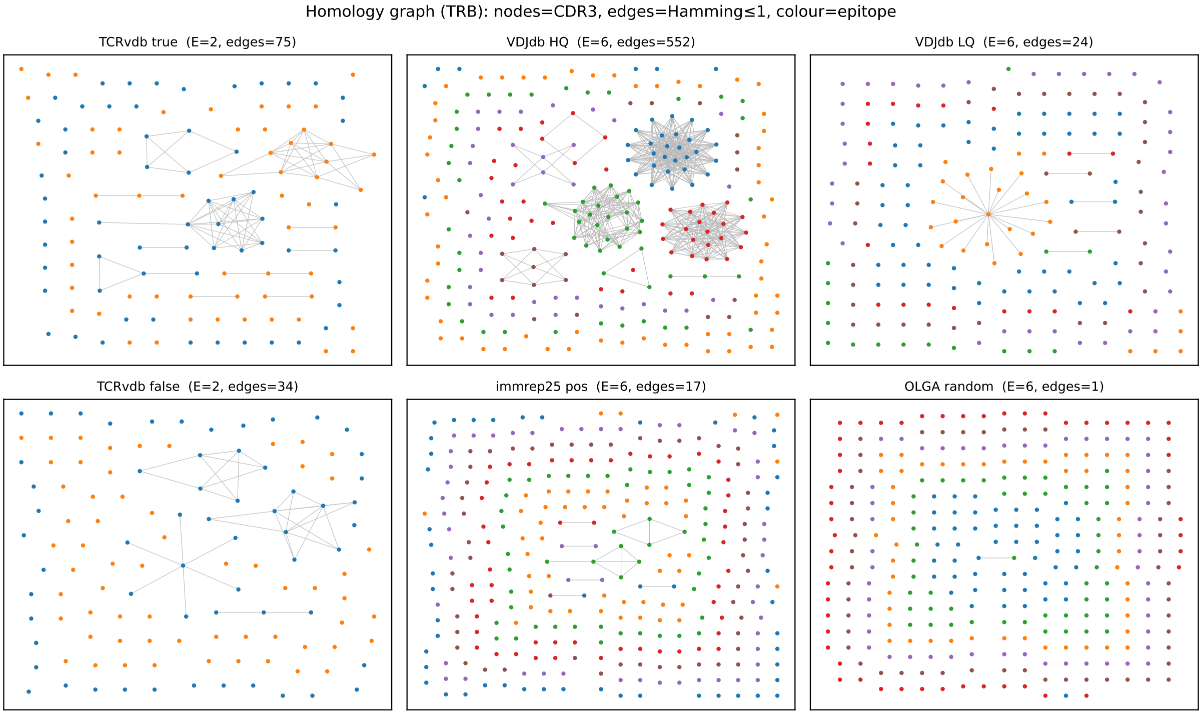

Figure 3: TRβ homology graph per cohort (equal sampling). Edge counts: VDJdb-HQ ∼550, TCRvdb-true 75, versus immrep25 ∼17 and OLGA ∼1.

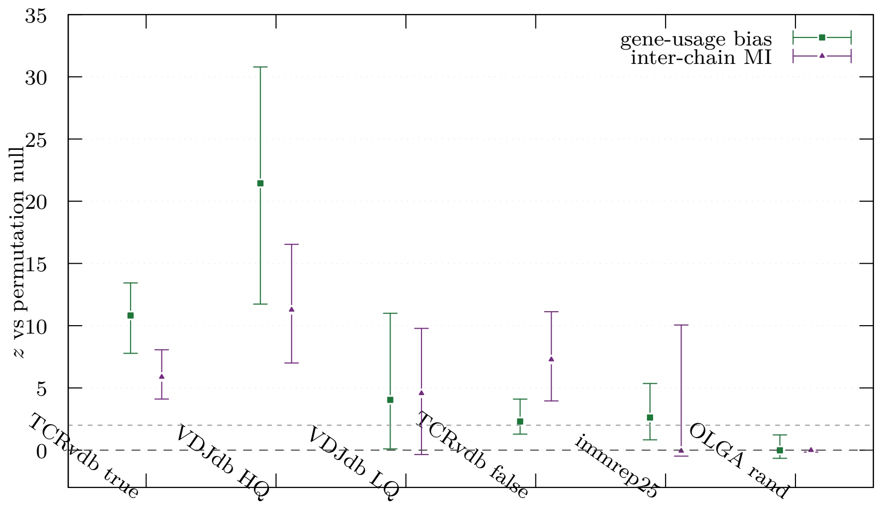

Figure 4: Pairing non-randomness z (median, 95% CI) per cohort: gene-usage bias (squares) and inter-chain mutual information (triangles). Dashed line z=0 (random), dotted z=2. immrep25 and OLGA sit at the random line on both.

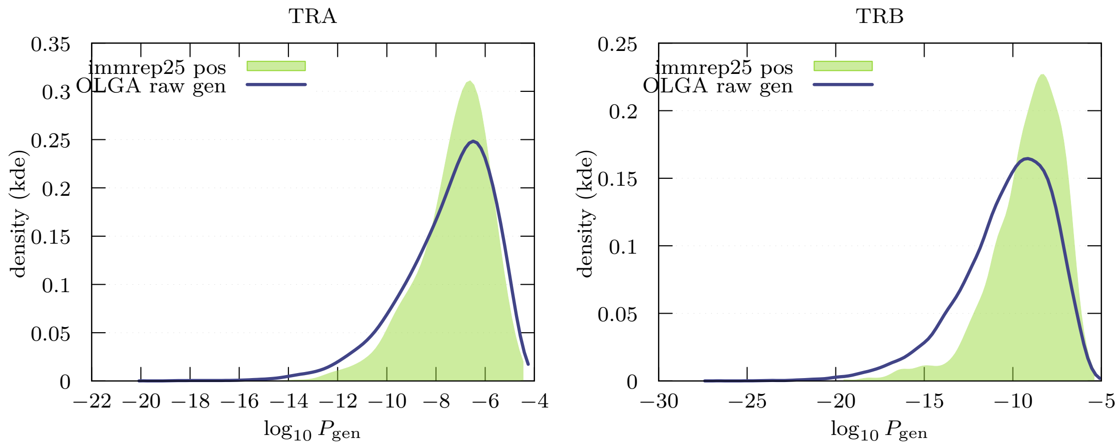

Figure 5: Per-chain $\log_{10} P_{\mathrm{gen}}$ (kernel density): immrep25 positives (filled) vs. raw OLGA generation (line). Positives skew to higher P_{gen} (public-leaning), motivating the P_{gen}-matched control.

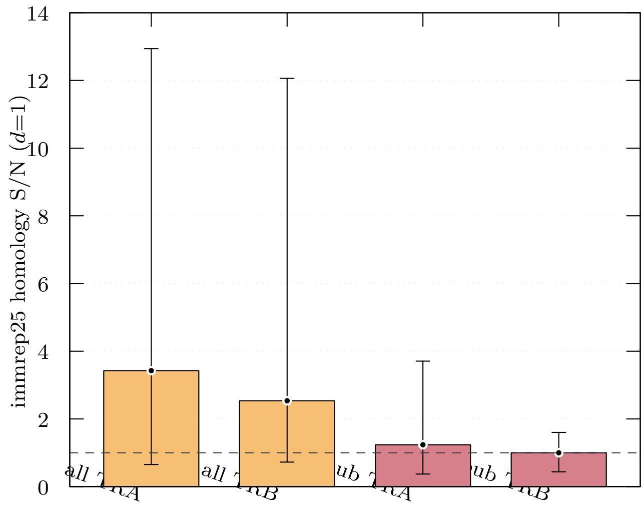

Figure 6: immrep25 homology S/N (d=1) for all positives vs. non-public positives (Hamming ≤ 1 to references removed). Removing public TCRs drives S/N to the noise floor (dashed line).

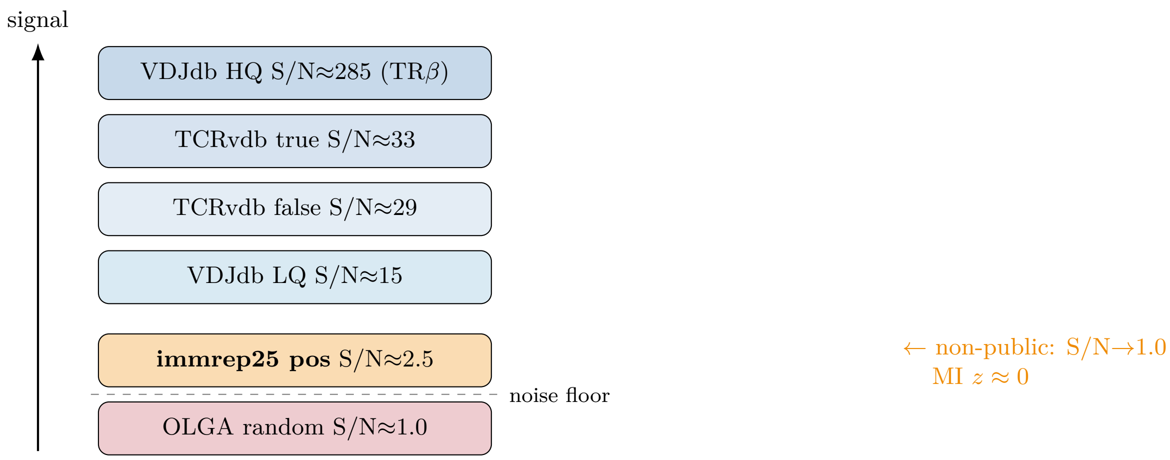

Figure 7: Signal-to-noise ladder (TRβ homology, matched design). The immrep25 positives sit one rung above the random floor; stripping public TCRs lowers them onto it.